

# iSAQB Module: Formal Methods - Logic

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# Logic

- Propositional Logic
- Predicate (First-Order) Logic
- Temporal Logic

Moreover:

- Model Theory vs.
- Proof Theory

# Propositional Logic

Propositional formulae  $F, G, \dots \in \mathcal{D}$

- *atomic formulae*  $A, B, C, \dots \in \mathcal{A} = \{A_1, A_2, A_3, \dots\}$
- *constant literals*  $\top, \perp$
- *negation*  $\neg A$
- *literals*  $L \in \mathcal{L} = \{\top, \perp, A_1, A_2, A_3, \dots, \neg A_1, \neg A_2, \neg A_3, \dots\}$
- *disjunction*  $(F \vee G)$
- *conjunction*  $(F \wedge G)$

# Notation

- *implication*  $(F \Rightarrow G) \equiv (\neg F \vee G)$
- *equivalence*  $(F \Leftrightarrow G) \equiv ((F \Rightarrow G) \wedge (G \Rightarrow F))$

## Exercise: General Negation

Negation is restricted to literals in this language. Why do you think that is? How could general negation be recovered?

# Semantics

- truth values  $\mathcal{T} = \{0, 1\}$
- assignment  $A : \mathcal{A} \rightarrow \mathcal{T}$

Extend assignment to *semantics/model*  $A' : \mathcal{D} \rightarrow \mathcal{T}$ .

1.  $A'(\top) = 1, A'(\perp) = 0$
2.  $A_i \in \mathcal{A} : A'(A_i) = A(A_i), A'(\neg A_i) = 1 - A(A_i)$
3.  $A'(F \wedge G) = \begin{cases} 1 & A'(F) = 1 \text{ and } A'(G) = 1 \\ 0 & \text{otherwise} \end{cases}$
4.  $A'(F \vee G) = \begin{cases} 1 & A'(F) = 1 \text{ or } A'(G) = 1 \\ 0 & \text{otherwise} \end{cases}$

# Validity

A formula  $F$  is *valid* if  $A'(F) = 1$  in every assignment  $A$  / model  $A'$ .

## Exercise: Semantics of Propositional Logic

Calculate the semantics of  $\neg((A \wedge B) \vee C)$  for the assignment  $A(A) = 1, A(B) = 1, A(C) = 0$ !

# Semantics Notation & Terminology

- $A \models F \equiv A'(F) = 1$ :  $A$  is a *model* for  $F$
- model exists:  $F$  is *satisfiable*
- no model exists:  $F$  is *unsatisfiable*
- all assignments are models:  $F$  is a *tautology*
- $A \models F$  iff  $A \models G$  for all  $A$ :  $F$  and  $G$  are *equivalent*

# Inference Calculus

- *axioms*
- *inference rules*
- *derivability/provability*

# Sequent Calculus

Multi-sets (*sequences*) of formulae  $\Gamma, \Pi, \Delta, \dots$

Axioms:  $\top, \Gamma$                        $\neg A_i, A_i, \Gamma$

Inference rules:

$$\frac{F, G, \Gamma}{F \vee G, \Gamma}$$

$$\frac{F, \Gamma \quad G, \Gamma}{F \wedge G, \Gamma}$$

# Derivability

Formula is derivable  $\vdash F$  iff a tree labelled with sequences exists where:

- root is the formula
- leaves are axioms
- each internal node is an instance of an inference rule

# Properties of a calculus

- *consistency* contradictions are not derivable
- *soundness* only tautologies are derivable
- *completeness* all tautologies are derivable
- *decidable* it is decidable whether a formula is derivable or not

*Theorem:* The sequent calculus all of the above.

## Exercise: Sequent Calculus Proofs

- Prove  $(A_1 \wedge A_2) \vee (\neg A_1 \vee \neg A_2)$  in the sequent calculus!
- What is a sensible strategy to construct a proof for a given formula?

# Classical vs. Intuitionistic Logic

In the Sequent Calculus,  $F \vee \neg F$  (*LEM*) and  $F \Leftrightarrow \neg\neg F$  (double negation) are derivable for any  $F$ .

This enables *proof by contradiction*.

This is problematic in some settings, as proofs by contradiction are not *constructive*.

# Constructive Proofs

**Theorem** There are irrational numbers  $a$  and  $b$  such that  $a^b$  is rational.

Exercise: What would you expect from a proof of this theorem?

# Non-Constructive Proof

**Theorem** There are irrational numbers  $a$  and  $b$  such that  $a^b$  is rational.

Exercise: What are two numbers  $a$  and  $b$  such that  $a^b$  is rational?

# Non-Constructive Proof

Consider  $c = \sqrt{2}$ .  $c^c$  is either rational, or not.

1. Assume  $c^c$  is rational. Then we can choose  $a = b = c$  and we're done.
2. Assume  $c^c$  is irrational. Then we can choose  $a = c^c = \sqrt{2}^{\sqrt{2}}$  and  $b = c$ . Both are irrational.

$$a^b = \left( \sqrt{2}^{\sqrt{2}} \right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^2 = 2$$

*Intuitionistic logic* would not admit this proof.

# Predicate Logic

- *variables*  $u, v, w, x, y, z, \dots \in \mathcal{V} = \{x_1, x_2, x_3, \dots\}$
- *function symbols*  $f, g, h, \dots \in \mathcal{F} = \{f_1, f_2, f_3, \dots\}$
- each function symbol has *arity*  $k \in \mathcal{N}$

*Constants*  $a, b, c, \dots$  are function symbols with arity 0.

*Terms*  $a, b, c, \dots, t_1, \dots, t_n$ :

1. Each variable is a term.
2.  $f(t_1, \dots, t_k)$  if  $f$  has arity  $k$

# Formulae

- *predicate symbols*  $P, Q, R, \dots \in \mathcal{P} = \{P_1, P_2, P_3, \dots\}$
- each predicate symbol has *arity*  $k \in \mathbb{N}$

Formulae  $F, G, \dots \in \mathcal{D}$ :

1. *constant literals*  $\top, \perp$
2. *atomic formulae*  $P(t_1, \dots, t_k)$  if  $P$  predicate symbol with arity  $k$
3.  $\neg F$
4.  $(F \vee G)$
5.  $(F \wedge G)$
6.  $(\exists x.F \quad \quad \quad)$  ( $\exists$  *existential quantifier*)
7.  $(\forall x.F \quad \quad \quad)$  ( $\forall$  *universal quantifier*)

# Free, Bound Variables

Each *occurrence* of a variable is either *bound* or *free*.

An occurrence of the variable  $x$  in the formula  $F$  is *bound* if  $x$  occurs within a subformula of  $F$  of the form  $\exists x.F$  or  $\forall x.F$ .

An occurrence of the variable  $x$  in the formula  $F$  is *free* if it is not bound.

Formula without occurrence of a free variable: *closed*.

# Exercise: Free and Bound Variables

Write a formal definition of free and bound variables!

# Semantics

*Structure S:*

- non-empty set  $U$  *universe*
- *function assignment* mapping each  $k$ -ary function symbol  $f \in \mathcal{F}$  to a  $k$ -ary function

$$\bar{f} : U \times \dots \times U \rightarrow U$$

- *predicate assignment* mapping each  $k$ -ary predicate  $P \in \mathcal{P}$  to a  $k$ -ary predicate

$$\bar{P} \in U \times \dots \times U$$

*Variable assignment*  $\beta : \mathcal{V} \rightarrow U$

Function, predicate, and variable assignments form an *interpretation*.

# Semantics

For a given structure  $S$  and variable assignment  $\beta$ , the *model*  $A_{S\beta} : \mathcal{D} \rightarrow \mathcal{T}$  of a formula is as follows:

1.  $A_{S\beta}(\top) = 1, A_{S\beta}(\perp) = 0$
2.  $x \in \mathcal{V} : A_{S\beta}(x) = \beta(x)$
3. 0-ary  $a \in \mathcal{F} : A_{S\beta}(a) = \bar{a}$
4.  $k$ -ary  $f \in \mathcal{F} : A_{S\beta}(f(t_1, \dots, t_k)) = \bar{f}(A_{S\beta}(t_1), \dots, A_{S\beta}(t_k))$
5.  $k$ -ary  $P \in \mathcal{P} : A_{S\beta}(P(t_1, \dots, t_k)) = \begin{cases} 1 & (A_{S\beta}(t_1), \dots, A_{S\beta}(t_k)) \in \bar{P} \\ 0 & \text{otherwise} \end{cases}$

# Semantics

$$6. A_{S\beta}(\neg F) = \begin{cases} 1 & A_{S\beta}(F) = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$7. A_{S\beta}(F \wedge G) = \begin{cases} 1 & A_{S\beta}(F) = 1 \text{ and } A_{S\beta}(G) = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$8. A_{S\beta}(F \vee G) = \begin{cases} 1 & A_{S\beta}(F) = 1 \text{ or } A_{S\beta}(G) = 1 \\ 0 & \text{otherwise} \end{cases}$$

# Semantics

$$\begin{aligned} 9. A_{S\beta}(\exists x_i.F) &= \begin{cases} 0 & A_{S\beta'}(F) = 0 \\ & \text{for each variable assignment } \beta' \\ & \text{with } \beta'(x_j) = \beta(x_j) \text{ for } j \neq i \\ 1 & \text{otherwise} \end{cases} \\ 10. A_{S\beta}(\forall x_i.F) &= \begin{cases} 1 & A_{S\beta'}(F) = 1 \\ & \text{for each variable assignment } \beta' \\ & \text{with } \beta'(x_j) = \beta(x_j) \text{ for } j \neq i \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

# Terminology

- *satisfiable, unsatisfiable, tautology* all as with propositional logic

# Sequent Calculus

Propositional calculus +

$$\frac{F[x \mapsto b], \exists x.F, \Gamma}{\exists x.F, \Gamma}$$

$$\frac{F[x \mapsto y], \Gamma}{\forall x.F, \Gamma} \text{ where } y \notin \text{free}(\forall x.F) \cup \text{free}(\Gamma)$$

## Exercise: Correctness of Lane-Change Assistant

Write a criterion for the correctness of a lane-change assistant in predicate logic.

A lane-change assistant first looks for a gap in traffic either on the left or right of the car. If it finds a gap, it will change to the lane of the gap.

# Linear Temporal Logic

Formulae of LTL  $\phi, \rho, \dots \in \mathcal{L}$ :

- $\top, \perp$
- *atomic propositions*  $A, B, C, \dots \in \mathcal{A} = \{A_1, A_2, A_3, \dots\}$
- *negation*  $\neg\phi$
- *disjunction*  $(\phi \vee \rho)$
- *conjunction*  $(\phi \wedge \rho)$
- *next*  $(\mathbf{X}\phi)$
- *until*  $(\phi \mathbf{U} \rho)$

# LTL Semantics

- *valuation*  $\sigma \in \mathcal{P}(\mathcal{A})$  true propositions
- *computation*  $\sigma = \sigma_0\sigma_1\sigma_2 \dots \in \mathcal{P}(\mathcal{A})^\omega$   
infinite sequence of valuations
- $\sigma^i = \sigma_i\sigma_{i+1}\sigma_{i+2} \dots$

# LTL Satisfaction Relation

$\sigma \models \phi$ :

- $\sigma \models \top$
- $\sigma \models A$  iff  $A \in \sigma_0$
- $\sigma \models \neg A$  iff  $\sigma \not\models A$
- $\sigma \models \phi \vee \rho$  iff  $\sigma \models \phi$  or  $\sigma \models \rho$
- $\sigma \models \phi \wedge \rho$  iff  $\sigma \models \phi$  and  $\sigma \models \rho$
- $\sigma \models \mathbf{X}\phi$  iff  $\sigma^1 \models \phi$
- $\sigma \models \phi \mathbf{U} \rho$  iff there is  $k \geq 0$  with  
     $\sigma^k \models \rho$  and  
     $\sigma^i \models \phi$  for all  $0 \leq i < k$

# LTL Syntactic Sugar

- *eventually*  $(\mathbf{F}\phi) = \top \mathbf{U} \phi$   
 $\sigma \models \mathbf{F}\phi$  iff there is  $k \geq 0$  with  $\sigma^k \models \phi$
- *always*  $(\mathbf{G}\phi) = \neg(\mathbf{F}\neg\phi)$   
 $\sigma \models \mathbf{G}\phi$  iff  $\sigma^k \models \phi$  for every  $k \geq 0$

# Exercise: LTL Tutor

Do a few exercises in the **LTL Tutor**!

<https://ltl-tutor.xyz/>

# Exercise: Vending Machine State Diagram

Consider a vending machine that:

- accepts a coin
- then allows selecting tea or coffee
- after dispensing the drink, goes back to the beginning

Draw a state diagram for the vending machine, and formulate LTL properties that a vending machine should fulfill.

# Kripke Structure

A *Kripke structure* is a tuple  $(W, \approx, w, I)$  where:

- $W$  set of *states*
- $\approx \in W \times W$  *transition relation*
- $w : W \rightarrow \mathcal{P}(\mathcal{A})$  maps states to valuations
- $I \subseteq W$  *initial states*

# Computation Structure

A Kripke structure is a *computation structure* if:

- $W$  is finite
- every  $s \in W$  has a successor  $t$  with  $s \leadsto t$ .

A *computation path* is an infinite sequence  $s_0 s_1 s_2 \dots$  with  $s_i \in W$  and  $s_i \leadsto s_{i+1}$  for all  $i$ .

# LTL Semantics / Computation Structure

Given an LTL formula  $\phi$  and a computation structure  $K = (W, \rightsquigarrow, w, I)$ , then

$$K \models \phi$$

$$\text{iff } \sigma \models \phi$$

for all  $\sigma$  where

$$\sigma_i = w(s_i) \text{ for some computation path } s_0 s_1 s_2 \dots \text{ in } K$$